Notes: Koudijs (JF, 2016)

The Boats That Did Not Sail: Asset Price Volatility in a Natural Experiment

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1 Big picture

- **Question.** What drives short-run stock-price volatility: public news, private information, liquidity trades, or sentiment? The paper studies a setting where public information flows can be observed directly and sometimes stop for several days.
- **Natural experiment.** In the 18th century, English securities were traded in Amsterdam, but relevant information about these securities came mainly from London. Official mail packet boats crossed the North Sea; adverse winds often delayed them, producing observable news blackouts in Amsterdam.
- **Core idea.** If no boat arrived, Amsterdam investors received no new public information from London. The market nevertheless remained open, so returns during no-boat periods reveal how much volatility can arise without fresh public news.
- **Main empirical fact.** Boat arrivals mattered a lot: returns were much more volatile when a boat arrived. But no-boat volatility was also large. The no-boat/boat variance ratio lies between roughly 0.40 and 0.58 across securities.
- **Main decomposition.** Public news explains a large share of volatility on boat-arrival days and around 40% of total volatility. Private information and local liquidity shocks explain much of the remaining volatility, especially on no-boat days.
- **Private information channel.** Boats carried not only newspapers and public news but also private letters and trading orders. Informed traders could spread trades over time, causing Amsterdam and London prices to move in the same direction even when no new public information had arrived in Amsterdam.
- **Liquidity channel.** The paper finds evidence consistent with limited risk-bearing capacity: uninformed order flow could move prices temporarily, and those price impacts could later reverse.
- **Important exception.** The British East India Company (EIC) has unusually high residual no-boat volatility. The paper interprets this as possibly due to liquidity trades concentrated in no-news periods because informed trading was especially severe around boat arrivals, or to sentiment and disagreement in the absence of objective news.
- **Economic takeaway.** Public news is important, but trading itself also creates substantial short-run price movements. Even in a market starved of public news, prices move because private information and liquidity shocks are revealed through trading.

2 Incremental contribution of the paper

- **From noisy news measurement to directly observed news flows.** Modern studies often struggle to identify what counts as relevant public news. This paper uses physical boat arrivals as an observable and plausibly exogenous measure of when London news reached Amsterdam.
- **A market-open news blackout.** Prior work studies periods when markets are closed or when news arrival is hard to measure. This paper studies a market that continues trading while public information from the main source is shut down.

- **A clean separation between news and trading.** Because Amsterdam traded on days with and without new London information, the paper can compare price movements after public-news arrivals with price movements generated by the trading process itself.
- **High-frequency historical data.** The paper hand-collects Amsterdam prices at three observations per week, links them to London prices, and combines them with detailed packet-boat arrival data.
- **Variance decomposition.** The paper does not stop at showing that no-news volatility exists. It decomposes return variance into public news, private information, and trading-cost or liquidity components.
- **Bridge between historical finance and market microstructure.** The setting is historical, but the mechanisms are modern: price discovery, informed trading, liquidity shocks, transaction costs, and imperfect arbitrage.
- **Evidence against a pure public-news view.** The paper shows that public information is essential, yet far from sufficient. A large fraction of short-run volatility is generated through private information and market-making or inventory effects.

3 Data and measurement

3.1 Historical market setting

- Amsterdam and London were major securities markets in the 18th century. Dutch investors held a substantial share of English securities, creating active Amsterdam trading in assets whose main information source was London.
- The market was sophisticated: there was a centralized exchange, active futures and options trading, shorting through futures, and a developed market for margin loans.
- The analysis focuses on periods of peace, September 1771 to December 1777 and September 1783 to March 1787. This restriction matters because it makes it more plausible that relevant information about English securities originated in London rather than elsewhere in Europe.

3.2 News-flow data

- London and the Dutch Republic were connected by mail packet boats sailing between Harwich and Hellevoetsluis.
- Boats were scheduled twice a week, but bad weather, especially eastern winds, delayed crossings. These delays created periods with no fresh public news from London in Amsterdam.
- Boat-arrival data come from Dutch and English newspapers. The paper also accounts for rare cases where news arrived through other channels, such as Ostend.
- The key measured treatment is whether a price observation in Amsterdam followed the arrival of a boat carrying London news.

3.3 Security-price data

- The main Amsterdam securities are the British East India Company (EIC), South Sea Company (SSC), Bank of England (BoE), and 3% and 4% annuities.
- Amsterdam prices are observed three times per week: Monday, Wednesday, and Friday. Returns are two- or three-day log percentage returns.
- Amsterdam prices refer to futures, so the paper converts them to spot-equivalent prices using a cost-of-carry adjustment.
- London prices are available at daily frequency for the EIC, BoE, and 3% annuities. These are the securities used in the cross-market response and private-information tests.

3.4 Boat and no-boat periods

- The paper separates Amsterdam returns into *boat returns*, when a boat carrying London news arrived, and *no-boat returns*, when no new London news reached Amsterdam.
- In the two sample periods, there are 681 between-boat periods. Of these, 383 are long enough to contain at least one no-boat return.
- The median gap between boats is three days, but the distribution has a fat right tail. The maximum gap is 21 days.

3.5 Outcome measures

- The main outcome is the variance of Amsterdam security returns, separately for boat and no-boat periods.
- The paper also studies covariances between Amsterdam returns and London returns before and after boat departures, autocovariances in Amsterdam returns, and cross-market return reversals.
- These covariance patterns are used to distinguish public news, private information, and local liquidity shocks.

4 Models and empirical specifications

4.1 Market response to news

The first validation exercise asks whether Amsterdam prices respond to price changes in London after a boat arrives. A simplified version of the core specification is

$$\Delta p_t^{AMS,boat} = \alpha_0 + \alpha_1 \Delta p_s^{LND} + \alpha_2 \Delta p_{s-2}^{AMS} + u_t,$$

where $\Delta p_t^{AMS,boat}$ is an Amsterdam return that reflects the arrival of a boat and Δp_s^{LND} is the London price change since the previous boat departure.

Interpretation: A large positive α_1 means that London-originating news moves Amsterdam prices when the boat arrives. The reverse regression, London responding to Amsterdam, is used to check the direction of information flow.

4.2 Boat versus no-boat variance comparison

The benchmark comparison is the variance ratio

$$\frac{\mathrm{var}(\Delta p^{AMS,\textit{no-boat}})}{\mathrm{var}(\Delta p^{AMS,\textit{boat}})}.$$

- If public news were the only source of short-run volatility, this ratio should be close to zero.
- If the trading process itself generates volatility, this ratio should be economically meaningful.
- The paper finds ratios between 0.40 and 0.58, depending on the security.

Interpretation: Public news clearly raises volatility, but no-news trading still produces substantial price movements.

4.3 Delayed response to public news

The paper tests whether no-boat volatility reflects a slow response to previously arrived public news. A simplified specification is

$$\Delta p_t^{AMS} = \alpha_0 + \alpha_1 \Delta p_{t-1}^{AMS,boat} + \alpha_2 \Delta p_{s-2}^{LND} + \varepsilon_t,$$

estimated for cases where a boat arrived in the previous period.

- A slow response to public news would imply positive continuation from London news.
- The paper finds little evidence of such momentum.

Interpretation: No-boat volatility is not mainly a delayed reaction to public news that arrived earlier.

4.4 Private information specification

The private-information test uses the idea that informed agents in both London and Amsterdam trade gradually on the same private signals. For boat returns:

$$\Delta p_t^{AMS,boat} = \alpha_0 + \alpha_1 \Delta p_s^{LND} + \alpha_2 \Delta p_{s-1}^{LND} + u_t.$$

For no-boat returns:

$$\Delta p_{t+d}^{AMS,\textit{no-boat}} = \beta_0 + \beta_1 \Delta p_s^{LND} + v_t.$$

Here Δp_s^{LND} captures contemporaneous London price movement after a boat has departed but before the next boat transmits that information to Amsterdam.

Interpretation: If Amsterdam returns covary with contemporaneous but as-yet unreported London returns, this suggests that the same private information is being incorporated into both markets through trading.

4.5 Variance decomposition

The paper formalizes the decomposition of Amsterdam return variance into public news, private information, and residual components. The core idea is summarized by equations of the form

$$\mathrm{var}(\Delta p_t^{AMS,news}) = \mathrm{cov}(\Delta p_t^{AMS,news}, \Delta p_{s-1}^{LND}) + \mathrm{cov}(\Delta p_t^{AMS,news}, \Delta p_s^{LND}) + \sigma^2,$$

and

$$\mathrm{var}(\Delta p_{t+d}^{AMS,no\text{-}news}) = \mathrm{cov}(\Delta p_{t+d}^{AMS,no\text{-}news}, \Delta p_s^{LND}) + \sigma^2.$$

Interpretation: Covariance with predeparture London returns identifies public news; covariance with postdeparture London returns identifies private information.

4.6 Trading costs and liquidity shocks

The paper also estimates transaction-cost and inventory-cost components. A simple return equation is

$$\Delta p_t = u_t + \lambda q_t,$$

where q_t is order flow and λ captures price impact or inventory compensation.

- Order-handling costs are estimated using Roll-style autocovariances.
- Inventory costs are estimated by simulating a model in which order flow responds slowly to mispricing.
- Negative autocovariance in returns is interpreted as evidence of temporary price impact from liquidity shocks.

Interpretation: Part of no-news volatility reflects temporary price pressure from local liquidity trades rather than information.

5 Identification and empirical design

5.1 What the design identifies

- The core comparison identifies how much Amsterdam prices move when London public news arrives versus when no London public news arrives.

- Cross-market covariance tests identify whether prices in Amsterdam move with London prices even when no direct information sharing is possible.
- Autocovariance and reversal tests identify temporary price impact from local liquidity trades.

Interpretation: The design identifies components of short-run volatility, not only the total amount of volatility.

5.2 Key identifying assumptions

- **London-originating information.** During the sample periods, material information about English securities mostly originated in London.
- **Boats as main channel.** Packet boats were the main channel for newspapers, newsletters, public prices, private letters, and trading orders from London.
- **Weather-driven delays.** Delays were caused by adverse winds and weather, not by stock-market conditions.
- **No alternative systematic news channel.** Other ships, carrier pigeons, and third-country news sources do not appear to drive the results.
- **Market remained open.** Amsterdam continued to trade during news blackouts, allowing the paper to observe returns in the absence of new public information.

5.3 Validation evidence

- Amsterdam responds strongly to London price changes after boats arrive, but London does not respond much to Amsterdam price changes during peace periods.
- Hasbrouck information shares indicate that most price discovery occurs in London during the main sample periods.
- English security returns in Amsterdam are not meaningfully correlated with Dutch security returns on no-boat days.
- Newspaper content linked to the EIC predicts EIC return volatility, showing that boats transmitted material news.
- There is little evidence that transitory London liquidity shocks were transmitted to Amsterdam.

Interpretation: The validation exercises support the central claim that boat arrivals capture the arrival of relevant London public news, not merely endogenous trading activity.

5.4 Main limitations of the design

- The setting is historical; the market is sophisticated but still differs from modern electronic markets.
- Volume data are mostly unavailable; the transaction evidence is based on a small sample.
- Boats transmitted both public news and private letters, so the paper must use covariance structure to separate public and private information.

- The EIC has a large residual no-boat component that cannot be cleanly separated into liquidity shocks, adverse-selection-driven trading concentration, sentiment, or disagreement.

6 Tables and figures

6.1 Table I and Figure 1: data structure and two motivating episodes

Read:

- Table I reports descriptive statistics for Amsterdam returns. During the two main sample periods, each security has 1,162 return observations, including 681 news returns and 481 no-news returns.
- The sample periods are less volatile than the full 1771-1787 period because the intermediate war years were excluded.
- Panel C shows that the median time between boats is three days, but some gaps are much longer, reaching 21 days.
- Figure 1 gives two concrete examples. In Panel A, London receives bad EIC news first and Amsterdam prices adjust only after the delayed boat arrives. In Panel B, Amsterdam prices move during a news blackout, illustrating no-news volatility.

Economic message: The paper's empirical design rests on observable variation in the arrival of London news. Figure 1 makes the identification intuitive: prices respond sharply when boats transmit news, but they can also move when boats do not arrive.

Table I

Descriptive Statistics

Panel A presents descriptive statistics for Amsterdam security returns for the two sample periods (1771 to 1777 and 1783 to 1787) as well as the entire 1771 to 1787 period. Two- or three-day log percentage returns are calculated as $\log(p_t/p_{t-1})$. Panel B provides descriptive statistics for Amsterdam security returns for nonoverlapping two-week periods, based on the data collected by Van Dillen (1931). The table includes a Brown-Forsythe test statistic on the equality of different return variances. Panel C summarizes between-boat periods for the two sample periods. Periods involving a no-boat return are long enough that there is at least one price change in the absence of news. *** and ** denote statistical significance at the 1% and 5% level, respectively.

	EIC	SSC	BoE	3% ann.	4% ann.
Panel A: Return Information 1771 to 1787 (two- or three-day returns)					
September 1771 to December 1777 and September 1783 to March 1787 (the two sample periods)					
Mean	0.056	0.031	0.032	0.042	0.033
Variance	0.528	0.264	0.188	0.330	0.237
N (total)	1,162	1,162	1,162	1,162	1,162
N (news returns)	681	681	681	681	681
N (no-news returns)	481	481	481	481	481
September 1771 to March 1787 (the two sample periods and the intermediate war period)					
Mean	0.051	0.034	0.040	0.035	0.044
Variance	0.618	0.443	0.292	0.523	0.396
N (total)	1,900	1,900	1,900	1,900	1,767
B-F statistic on equality return variance	(12.4)***	(51.9)***	(52.8)***	(63.8)***	(79.4)***
Panel B: Return Information 1723 to 1794 (nonoverlapping two-week returns)					
September 1771 to December 1777 and September 1783 to March 1787 (the two sample periods)					
Mean	0.079	−0.003	0.019	N/A	
Variance	5.173	1.920	2.448		
N	133	133	133		
November 1723 to December 1794 (the full period for which data are available, with gaps)					
Mean	0.037	0.047	0.007	N/A	
Variance	3.876	2.236	2.534		
N	957	957	957		
B-F statistic on equality return variance	(4.24)**	(0.039)	(0.046)		

Panel C: Between-boat periods (September 1771 to December 1777 and September 1783 to March 1787)					
Number of between-boat periods	681				
Those involving a no-boat return	383				
	min	5th perc.	median	95th perc.	max
Length between boat periods (days)	1	2	3	7	21
Number of price observations per period	0	1	2	3	8

Table I: Descriptive Statistics

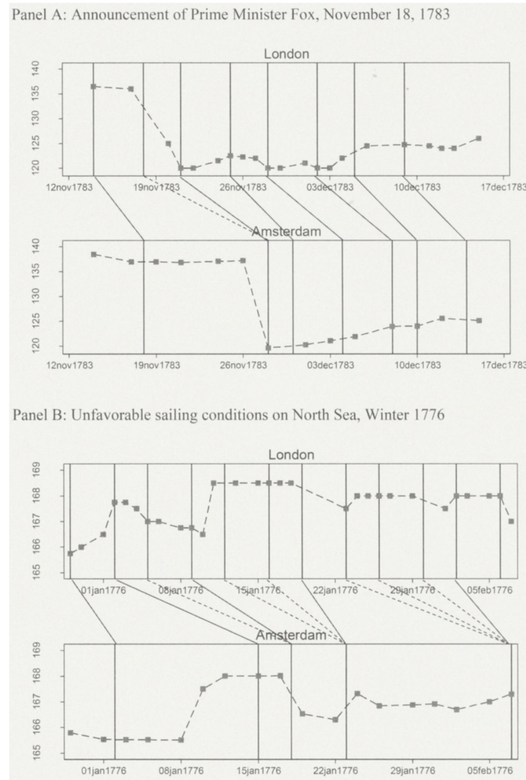

Figure 1. Examples. The figure presents examples of news transmission and price movements for EIC stock. Vertical/diagonal lines indicate the flow of news between London and Amsterdam, solid lines indicate the most recent news transmission, dashed lines indicate stale news, and squares

Figure 1: Examples

6.2 Figure 2 and Table II: return timing and the London-to-Amsterdam response

Read:

- Figure 2 defines the timing of returns across the two markets and clarifies the distinction between boat returns and no-boat returns.
- Table II shows that Amsterdam responds strongly to London price changes after a boat arrives: coefficients are about 0.38 for EIC, 0.43 for BoE, and 0.48 for 3% annuities.
- The reverse relationship is much weaker. London responses to Amsterdam returns are small, with near-zero adjusted R^2 values.

Economic message: Information mainly flows from London to Amsterdam, not the other way around, which validates the paper's focus on London-originating news.

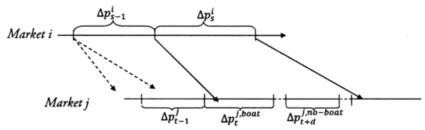

Figure 2. Structure of return data. This figure illustrates the structure of the return data. $i, j \in \{LND, AMS\}$, with $i \neq j$, t indexes returns over two or three days (Monday-to-Wednesday, Wednesday-to-Friday, or Friday-to-Monday), s indexes periods in the other market between two subsequent boat departures. $\Delta p_t^{j,boat}$ reflects the arrival of a boat that transmits between-boat return Δp^i_{s-1}. If the period between the arrival of two boats is long enough, there will be returns that take place in the absence of a boat arrival, $\Delta p_{t+d}^{j,no=boat}$. There are at most seven no-boat returns per period; $d \in [1, 7]$ (see Table I). Δp^i_{s-1} is the contemporaneous return in the other market. Δp^j_{t-1} is the preceding return in the own market, which may or may not reflect the arrival of a boat.

Figure 2: Structure of Return Data

Table II

Response to News, Amsterdam and London

This table presents estimates of the response of news returns in Amsterdam (AMS) and London (LND) to returns observed in the other market: $\Delta p_t^{AMS.boat} = \alpha_0 + \alpha_1 \Delta p_{s-1}^{LND} + \alpha_2 \Delta p_{s-2}^{AMS} + u_t$ and $\Delta p_t^{LND.boat} = \beta_0 + \beta_1 \Delta p_{s-1}^{AMS} + \beta_2 \Delta p_{s-2}^{LND} + v_t$. $\Delta p_t^{AMS.boat}$ and $\Delta p_t^{LND.boat}$ reflect the arrival of a boat from the other market and are calculated as percentage log returns over two- or three-day periods. Returns observed in the other market, Δp_{s-1}^{LND} and Δp_{s-1}^{AMS}, are defined as price changes between boat departures. Δp_{s-2}^{LND} and Δp_{s-2}^{AMS} are past London and Amsterdam returns. Robust bootstrapped (1,000 replications) standard errors are reported in parentheses. Sample periods: September 1771 to December 1777 and September 17 to March 1787. ***, **, * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	AMS (response to LND) $\Delta p_t^{AMS.boat}$			LND (response to AMS) $\Delta p_t^{LND.boat}$		
	EIC	BoE	3% Ann.	EIC	BoE	3% Ann.
Observed LND return	0.380	0.425	0.483			
(Δp_{s-1}^{LND})	(0.041)***	(0.058)***	(0.074)***			
Past AMS return	−0.021	−0.004	−0.040			
(Δp_{s-2}^{AMS})	(0.038)	(0.003)	(0.038)			
Observed AMS return				0.056	0.053	0.064
(Δp_{s-1}^{AMS})				(0.033)*	(0.038)	(0.033)*
Past LND return				−0.026	−0.068	−0.056
(Δp_{s-2}^{LND})				(0.035)	(0.052)	(0.036)
Constant	0.054	0.018	0.037	0.006	0.019	0.016
	(0.030)*	(0.015)	(0.021)*	(0.032)	(0.019)	(0.017)
N	594	602	621	636	629	756
Adj. R^2	0.21	0.26	0.24	0.00	0.00	0.01

Table II: Response to News, Amsterdam and London

6.3 Tables III and IV: direction of news and Dutch-market controls

Read:

- Table III reports Hasbrouck information shares. During the peace sample, the midpoint share attributed to London is about 0.93 for EIC, 0.92 for BoE, and 0.99 for 3% annuities.
- During the war period, the London share falls substantially, showing why the paper excludes that period from the main analysis.
- Table IV shows that no-news returns on English securities in Amsterdam are barely correlated with returns on Dutch securities such as VOC, WIC, or bank money.

Economic message: The validation evidence supports the assumption that, during the sample periods, relevant information about English securities comes mainly from London rather than Amsterdam or broader Dutch-market conditions.

Table III

Hasbrouck Information Shares

This table reports Hasbrouck information shares based on VECM models that use prices in Amsterdam and London three days a week (Monday, Wednesday, and Friday). The VECM models are estimated using five lags; each lag represents two or three days. The reported information shares are for different Cholesky orderings (either London or Amsterdam first) and the simple mean of the two. Panel A presents results for the two sample periods that are characterized by peace. Panel B presents results for the intermediate period of international conflict.

	Ordering		
	LND first	AMS first	Midpoint
Panel A: September 1771 to December 1777 and September 1783 to March 1787			
EIC	0.998	0.867	0.933
BoE	0.992	0.854	0.923
3% Ann.	0.999	0.973	0.986
Obs.	1,498		
Panel B: January 1778 to August 1783			
EIC	0.799	0.552	0.676
BoE	0.730	0.424	0.577
3% Ann.	0.864	0.647	0.756

Table III: Hasbrouck Information Shares

Table IV

Correlation between English and Dutch Security Returns in Amsterdam

This table presents regressions of English security returns on Dutch security returns in Amsterdam. Returns are for periods without news from London only. VOC: Dutch East India Company; WIC: Dutch West Indies Company; and Bank money (agio): return on deposit money at the Bank of Amsterdam relative to cash. Percentage log returns are calculated over two- or three-day periods. Sample periods: September 1771 to December 1777 and September 1783 to March 1787. ** denotes statistical significance at the 5% level.

		$\Delta p_{t+d}^{AMS,no=boat}$				
		EIC	SSC	BoE	3% Ann.	4% Ann
$\Delta p_t^{AMS,no=news}$	VOC	0.078	0.061	0.078	0.033	0.061
		(0.057)	(0.043)	(0.039)**	(0.045)	(0.041)
	WIC	0.031	0.001	−0.005	−0.006	−0.005
		(0.024)	(0.021)	(0.018)	(0.023)	(0.024)
	Bank money (agio)	0.290	0.171	−0.021	−0.137	−0.071
		(0.326)	(0.284)	(0.222)	(0.269)	(0.321)
Constant		0.013	0.015	0.003	0.002	0.014
		(0.024)	(0.020)	(0.017)	(0.021)	(0.018)

Table IV: Correlation between English and Dutch Security Returns in Amsterdam

6.4 Tables V and VI: news content and EIC reversals

Read:

- Table V links squared EIC news returns to capitalized newspaper keywords in the *Leydse Courant*. Multiple references to the EIC, financial terms, and especially Cape-related EIC news are associated with higher EIC volatility.
- This shows that packet boats transmitted economically meaningful information, not just generic news.
- Table VI studies whether EIC returns revert within Amsterdam, across markets, or within London. The EIC shows little evidence of negative return reversals in the main predictive regressions.

Economic message: Boat arrivals carried real news. At the same time, EIC price dynamics are special: they do not show the same clean liquidity-shock reversal pattern as the other securities.

Table V

The Impact of Different News Categories on EIC Returns

This table presents regression results relating squared Amsterdam EIC news returns to the publication of relevant capitalized words in articles with news from England published in the *Leydse Courant* between September 1771 and December 1777. *EIC*: East India Company; *Finance*: financial terminology (excludes "stocks"); *India*: reference to regions, towns, and/or persons in the British East Indies; *Cape*: references to the Cape (of Good Hope) Colony; *Background*: words signaling background information (Section V.B of the Internet Appendix provides further details). News related to the EIC is captured by two dummy variables: *EIC (1)* takes a value of one if the EIC is mentioned once in an article. *EIC (>1)* takes a value of one if the EIC is mentioned at least twice. The other news variables are also included as dummies, where a dummy takes a value of one if a specific category is referenced at least once. The "Obs." column indicates for how many observations a certain (combination of) dummy(ies) is turned on. Robust bootstrapped (1,000 replications) standard errors are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	$(\Delta p_t^{AMS.boat})^2$						
	(1)	(2)	(3)	(4)	(5)	(6)	Obs.
EIC (1)	–0.023	0.029	–0.022	0.077	0.015	0.095	101
	(0.183)	(0.186)	(0.202)	(0.184)	(0.225)	(0.241)	
EIC (> 1)	1.487	1.063	1.462	1.040	1.701	1.434	28
	(0.843)*	(0.802)	(0.961)	(0.838)	(0.927)*	(1.151)	
Finance	1.133	0.152	1.128	0.115	1.088	0.122	15
	(0.659)*	(0.501)	(0.724)	(0.398)	(0.669)	(0.482)	
India	–0.243	–0.349	–0.292	–0.485	–0.277	–0.334	32
	(0.260)	(0.238)	(0.145)**	(0.215)**	(0.274)	(0.142)**	
Cape	0.916	0.162	0.907	–0.449	0.878	–0.457	11
	(0.915)	(0.591)	(0.944)	(0.131)***	(0.886)	(0.119)***	
Background	–0.388	–0.348	–0.386	–0.346	–0.297	–0.257	98
	(0.130)***	(0.120)***	(0.126)***	(0.123)***	(0.145)**	(0.142)*	
EIC (1)*		0.151				0.156	5
Finance		(0.922)				(0.910)	
EIC (> 1)*		5.355				–1.176	3
Finance		(3.537)				(1.234)	
*EIC (1)*India*			0.038			–0.029	18
			(0.270)			(0.276)	
*EIC (> 1)*India*			0.190			–1.141	5
			(1.717)			(1.439)	
*EIC (1)*Cape*				–0.322		–0.311	3
				(0.212)		(0.251)	
*EIC (> 1)*Cape*				9.181		10.480	2
				(1.720)***		(1.494)***	
EIC (1)*					–0.166	–0.250	16
Background					(0.266)	(0.252)	
EIC (> 1)*					–1.806	–1.612	3
Background					(0.941)*	(1.164)	
Constant	0.665	0.689	0.667	0.702	0.649	0.681	
	(0.102)***	(0.099)***	(0.103)***	(0.100)***	(0.107)***	(0.102)***	
Obs.	512	512	512	512	512	512	
Adj. R^2	0.051	0.077	0.047	0.111	0.051	0.106	
Adj. R^2 using $\lvert\Delta p_t^{AMS.boat}\rvert$	0.045	0.063	0.041	0.083	0.046	0.077	

Table V: The Impact of Different News Categories on EIC Returns

Table VI

Predictive Regressions: EIC

This table presents results on whether, and over what horizons, returns revert. Panel A examines reversals within the Amsterdam market: $\Delta p_{t+T}^{AMS} = \alpha_0 + \alpha_1 \Delta p_t^{AMS} + u_t$, where Δp_{t+T}^{AMS} is the Amsterdam return from t to T, and T lies between two or three days and four weeks. Δp_t^{AMS} is the usual Amsterdam two- or three-day return. Panel B focuses on reversals across markets: $\Delta p_{t+T}^{AMS} = \beta_0 + \beta_1 \Delta p_{s-1}^{LND} + v_t$ for $B_t = 1$, where Δp_{t+T}^{AMS} is defined as before, $B_t = 1$ indicates that a boat arrived in t, and Δp_{s-1}^{LND} is the London return observed during that period. See Figure 2 for more details. Note that Δp_{t+T}^{AMS} does not capture the Amsterdam response to the news from London. Panel C examines return reversals within the London market: $\Delta p_{t+T}^{LND} = \gamma_0 + \gamma_1 \Delta p_t^{LND} + w_t$, where Δp_t^{LND} is the two- or three-day return in London and Δp_{t+T}^{LND} is the return in London between t and T. Robust bootstrapped (1,000 replications) standard errors are presented in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	2/3 days	4/5 days	1 week	2 weeks	3 weeks	4 weeks
	Panel A: Future Amsterdam EIC Returns (Δp_{t+T}^{AMS})					
Current Amsterdam EIC returns (Δp_t^{AMS})	0.009	0.006	0.005	0.035	0.039	0.088
	(0.034)	(0.052)	(0.056)	(0.072)	(0.092)	(0.0103)
Constant	0.032	0.063	0.092	0.172	0.253	0.339
	(0.018)*	(0.026)*	(0.032)***	(0.046)***	(0.058)***	(0.068)***
N	1,540	1,535	1,530	1,515	1,500	1,485
Adj. R^2	0.00	0.00	0.00	0.00	0.00	0.00
	Panel B: Future Amsterdam EIC Returns (Δp_{t+T}^{AMS})					
London news EIC returns (Δp_{s-1}^{LND})	0.041	0.028	0.076	0.054	0.073	0.178
	(0.043)	(0.060)	(0.060)	(0.076)	(0.103)	(0.115)
Constant	–0.012	0.005	0.060	0.125	0.153	0.207
	(0.028)	(0.039)	(0.048)	(0.067)*	(0.087)*	(0.102)**
N	737	736	735	731	724	722
Adj. R^2	0.00	0.00	0.00	0.00	0.00	0.00
	Panel C: Future London EIC Returns (Δp_{t+T}^{LND})					
Current London EIC returns (Δp_t^{LND})	0.020	–0.011	0.025	0.042	0.063	0.146
	(0.046)	(0.062)	(0.081)	(0.082)	(0.099)	(0.104)
Constant	0.027	0.042	0.053	0.106	0.157	0.199
	(0.023)	(0.032)	(0.039)	(0.055)*	(0.067)**	(0.077)***
N	1,167	1,302	1,322	1,335	1,336	1,336
Adj. R^2	0.00	0.00	0.00	0.00	0.00	0.00

Table VI: Predictive Regressions, EIC

6.5 Tables VII and VIII: return reversals for BoE and 3% annuities

Read:

- Tables VII and VIII repeat the predictive-reversal analysis for BoE and 3% annuities.
- For BoE and especially 3% annuities, there is evidence of short-horizon negative predictability within Amsterdam, consistent with temporary price impact from local liquidity shocks.
- Cross-market reversals are not negative, which weakens the interpretation that London liquidity shocks were transmitted to Amsterdam.

Economic message: Liquidity shocks appear to be local. Imperfect arbitrage and communication delays prevented temporary price pressure in one market from being immediately neutralized by the other market.

Table VII

Predictive Regressions: BoE

This table presents results on whether, and over what horizons, returns revert. Panel A examines reversals within the Amsterdam market: $\Delta p_{t+T}^{AMS} = \alpha_0 + \alpha_1 \Delta p_t^{AMS} + u_t$, where Δp_{t+T}^{AMS} is the Amsterdam return from t to T, and T lies between two or three days and four weeks. Δp_t^{AMS} is the usual Amsterdam two- or three-day return. Panel B focuses on reversals across markets: $\Delta p_{t+T}^{AMS} = \beta_0 + \beta_1 \Delta p_{s-1}^{LND} + v_t$ for $B_t = 1$, where Δp_{t+T}^{AMS} is defined as before, $B_t = 1$ indicates that a boat arrived in t, and Δp_{s-1}^{LND} is the London return observed during that period. See Figure 2 for more details. Note that Δp_{t+T}^{AMS} does not capture the Amsterdam response to the news from London. Panel C examines return reversals within the London market: $\Delta p_{t+T}^{LND} = \gamma_0 + \gamma_1 \Delta p_t^{LND} + w_t$, where Δp_t^{LND} is the two- or three-day return in London and Δp_{t+T}^{LND} is the return in London between t and T. Robust bootstrapped (1,000 replications) standard errors are presented in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	2/3 days	4/5 days	1 week	2 weeks	3 weeks	4 weeks
	Panel (A): Future Amsterdam BoE Returns (Δp_{t+T}^{AMS})					
Current Amsterdam	–0.049	–0.086	–0.049	0.009	0.032	0.154
BoE returns (Δp_t^{AMS})	(0.036)	(0.044)**	(0.054)	(0.077)	(0.096)	(0.118)
Constant	0.027	0.056	0.079	0.155	0.233	0.315
	(0.012)**	(0.016)***	(0.020)***	(0.028)***	(0.036)***	(0.043)***
N	1,540	1,535	1,530	1,515	1,500	1,485
Adj. R^2	0.00	0.00	0.00	0.00	0.00	0.00
	Panel (B): Future Amsterdam BoE Returns (Δp_{t+T}^{AMS})					
London news BoE	0.021	0.098	0.082	0.208	0.216	0.357
returns (Δp_{s-1}^{LND})	(0.057)	(0.078)	(0.080)	(0.102)**	(0.130)	(0.157)**
Constant	0.031	0.061	0.074	0.146	0.253	0.315
	(0.017)*	(0.023)***	(0.028)***	(0.039)***	(0.051)***	(0.059)***
N	707	707	705	699	692	689
Adj. R^2	0.00	0.01	0.00	0.01	0.01	0.02
	Panel (C): Future London BoE Returns (Δp_{t+T}^{LND})					
Current London BoE	–0.049	–0.106	–0.038	0.049	0.086	0.207
returns (Δp_t^{LND})	(0.045)	(0.054)**	(0.069)	(0.087)	(0.106)	(0.118)
Constant	0.029	0.054	0.081	0.165	0.235	0.311
	(0.014)**	(0.018)***	(0.021)***	(0.030)***	(0.038)***	(0.045)***
N	1,156	1,319	1,339	1,344	1,344	1,344
Adi. R^2	0.00	0.01	0.00	0.00	0.00	0.00

Table VII: Predictive Regressions, BoE

Table VIII

Predictive Regressions: 3 % Annuities

This table presents results on whether, and over what horizons, returns revert. Panel A examines reversals within the Amsterdam market: $\Delta p_{t+T}^{AMS} = \alpha_0 + \alpha_1 \Delta p_t^{AMS} + u_t$, where Δp_{t+T}^{AMS} is the Amsterdam return from t to T, and T lies between two or three days and four weeks. Δp_t^{AMS} is the usual Amsterdam two- or three-day return. Panel B focuses on reversals across markets: $\Delta p_{t+T}^{AMS} = \beta_0 + \beta_1 \Delta p_{s-1}^{LND} + v_t$ for $B_t = 1$, where Δp_{t+T}^{AMS} is defined as before, $B_t = 1$ indicates that a boat arrived in t, and Δp_{s-1}^{LND} is the London return observed during that period. See Figure 2 for more details. Note that Δp_{t+T}^{AMS} does not capture the Amsterdam response to the news from London. Panel C examines return reversals within the London market: $\Delta p_{t+T}^{LND} = \gamma_0 + \gamma_1 \Delta p_t^{LND} + w_t$, where Δp_t^{LND} is the two- or three-day return in London and Δp_{t+T}^{LND} is the return in London between t and T. Robust bootstrapped (1,000 replications) standard errors are presented in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	2/3 days	4/5 days	1 week	2 weeks	3 weeks	4 weeks
	Panel (A): Future Amsterdam 3% Ann. Returns (Δp_{t+T}^{AMS})					
Current Amsterdam 3%	–0.125	–0.189	–0.197	–0.164	–0.152	0.020
Ann. returns (Δp_t^{AMS})	(0.035)***	(0.049)***	(0.062)***	(0.080)*	(0.100)	(0.121)
Constant	0.033	0.064	0.093	0.169	0.251	0.332
	(0.015)**	(0.020)***	(0.023)***	(0.032)***	(0.039)***	(0.046)***
N	1,540	1,535	1,530	1,515	1,500	1,485
Adj. R^2	0.02	0.02	0.02	0.01	0.00	0.00
	Panel (B): Future Amsterdam 3% Ann. Returns (Δp_{t+T}^{AMS})					
London news 3%	0.030	0.107	0.097	0.199	0.213	0.431
Ann. returns (Δp_{s-1}^{LND})	(0.048)	(0.056)*	(0.078)	(0.092)**	(0.115)*	(0.133)***
Constant	0.007	0.041	0.068	0.140	0.245	0.310
	(0.019)	(0.025)	(0.029)**	(0.039)***	(0.050)***	(0.058)***
N	865	864	860	852	842	838
Adj. R^2	0.00	0.01	0.01	0.01	0.01	0.03
	Panel (C): Future London 3% Ann. Returns (Δp_{t+T}^{LND})					
Current London 3% Ann.	–0.031	–0.083	–0.059	0.035	0.064	0.191
returns (Δp_t^{LND})	(0.039)	(0.052)	(0.068)	(0.099)	(0.113)	(0.125)
Constant	0.026	0.056	0.075	0.151	0.233	0.315
	(0.013)**	(0.018)***	(0.022)***	(0.031)***	(0.039)***	(0.046)***
N	1,516	1,537	1,539	1,539	1,539	1,539
Adj. R^2	0.00	0.00	0.00	0.00	0.00	0.00

Table VIII: Predictive Regressions, 3% Annuities

6.6 Tables IX and X: benchmark variance and road map of mechanisms

Read:

- Table IX reports the core benchmark: return variance is higher on boat days for every security, but no-boat variance remains large.
- The no-boat/boat variance ratio ranges from 0.397 for EIC to 0.575 for BoE.
- Table X summarizes the candidate mechanisms. Public news from London affects only boat returns. Private information from London can affect both boat and no-boat returns. Local liquidity shocks can affect returns in Amsterdam.

Economic message: Public news is important, but no-news volatility is too large to ignore. The paper's main task is to explain that no-news volatility without collapsing it into public-news effects.

Table IX

Benchmark Results

This table presents descriptive statistics for Amsterdam percentage log returns over two- or three-day periods (denoted t) with or without a boat arrival ($B_t = 1$ or $B_t = 0$). Sample periods: September 1771 to December 1777 and September 1783 to March 1787. The equality of the variance of boat and no-boat returns is tested using a Brown-Forsythe (B-F) test ($H_0 : ratio = 1$). *** indicates statistical significance at the 1% level.

		Δp_t^{AMS}				
		EIC	SSC	BoE	3% Ann.	4% Ann.
Mean	Boat ($B_t = 1$)	0.084	0.037	0.043	0.062	0.041
	No-boat ($B_t = 0$)	0.018	0.023	0.017	0.012	0.022
	(t-statistic)	(1.526)	(0.456)	(1.032)	(1.350)	(0.635)
Variance	Boat ($B_t = 1$)	0.703	0.325	0.229	0.425	0.299
	No-boat ($B_t = 0$)	0.279	0.179	0.131	0.194	0.150
	(B-F statistic)	(42.3)***	(20.8)***	(24.8)***	(22.9)***	(30.4)***
Skewness	Boat ($B_t = 1$)	0.189	−0.030	0.169	0.441	−0.505
	No-boat ($B_t = 0$)	0.030	0.649	0.523	0.135	1.371
Kurtosis	Boat ($B_t = 1$)	7.52	8.52	7.99	11.38	10.40
	No-boat ($B_t = 0$)	8.68	7.88	10.95	8.22	10.87
% zero	Boat ($B_t = 1$)	13.5	38.0	22.8	29.5	50.6
	No-boat ($B_t = 0$)	27.0	54.7	38.3	41.0	65.3
Obs.	Boat ($B_t = 1$)	681	681	681	681	680
	No-boat ($B_t = 0$)	481	481	481	481	481
$\frac{var(\Delta p_{t+d}^{AMS.no-boat})}{var(\Delta p_t^{AMS.boat})}$		0.397	0.550	0.575	0.457	0.499

Table IX: Benchmark Results

Table X

Summary of Potential Explanations for Price Movements in Amsterdam

This table summarizes the potential factors influencing stock prices in Amsterdam during periods with and without boat arrivals, making a distinction between factors originating in London or Amsterdam. "Y/N": explanation relevant (yes/no) according to empirical evidence. The table reports that there is no evidence that liquidity or sentiment shocks from London affected prices in Amsterdam. This statement is based on the assumption that potential liquidity and sentiments shocks should, noticeably, revert at some time in the future.

Explanation:		Mechanism:	Boat Returns	No-Boat Returns
Public news	from London	Announcements	Y	N
	from Amsterdam	" "	N	N
Private information	from London	Trading activity	Y	Y
	from Amsterdam	" "	N	N
Liquidity shocks	from London	" "	N	N
	from Amsterdam	" "	Y (except EIC)	Y (except EIC?)
Sentiment shocks	from London	" "	N	N
	from Amsterdam	" "	N	N (except EIC?)

Table X: Summary of Potential Explanations for Price Movements in Amsterdam

6.7 Tables XI and XII: delayed public news versus private information

Read:

- Table XI tests whether returns show momentum after previous boat arrivals. There is little evidence that no-boat volatility is simply a delayed response to public news.
- Table XII shows that Amsterdam returns covary with contemporaneous London returns that have not yet been publicly transmitted to Amsterdam.
- The postdeparture London return is significant for both Amsterdam boat returns and Amsterdam no-boat returns.

Economic message: Private information is a major source of price movement. Prices can move in both markets in the same direction even before public information has crossed the North Sea.

Table XI

Momentum

This table presents estimates for

$$\Delta p_t^{AMS} = \alpha_0 + \alpha_1 \Delta p_{t-1}^{AMS.boat} + \alpha_2 \Delta p_{s-2}^{LND} + \varepsilon_t \text{ for } B_{t-1} = 1,$$

where $B_{t-1} = 1$ indicates that a boat arrived in period $t-1$, Δp_t^{AMS} is the current Amsterdam return, which can take place in the presence or absence of a boat, $\Delta p_{t-1}^{AMS.boat}$ is the previous period's Amsterdam boat return, and Δp_{s-2}^{LND} is the London news return observed in the previous period. α_1 measures the reversal of Amsterdam returns, and α_2 measures the continuation of London news returns. Robust bootstrapped (1,000 replications) standard errors are reported in parentheses. * indicates statistical significance at the 10% level.

	Current Amsterdam Return: Δp_t^{AMS}		
	EIC	BoE	3% ann.
Amsterdam news return, previous period	0.059	–0.055	–0.081
$(B_{t-1} = 1)$: $\Delta B_{t-1}^{AMS,news}$	(0.053)	(0.064)	(0.057)
London news return, observed in previous	–0.003	–0.038	–0.007
period $(B_{t-1} = 1)$: Δp_{s-2}^{LND}	(0.042)	(0.064)	(0.065)
Constant	0.009	0.032	0.031
	(0.029)	(0.017)*	(0.020)
N	623	634	655
Adj. R^2	0.00	0.01	0.01

Table XI: Momentum

Table XII

Private Information: News and No-News Returns

This table presents estimates of the impact of private information on security returns in Amsterdam on days with and without a boat arrival: $\Delta p_t^{AMS.boat} = \alpha_0 + \alpha_1 \Delta p_s^{LND} + \alpha_2 \Delta p_{s-1}^{LND} + u_t$ for $B_t = 1$ and $\Delta p_{t+d}^{AMS.no=boat} = \beta_0 + \beta_1 \Delta p_s^{LND} + v_t$ for $B_t = 0$. Returns are defined in Figure 2. London returns Δp_s^{LND} are defined as the price change after the departure of a boat (up to the departure of the next boat) and capture contemporaneous (but as-of-yet) unreported returns. For comparison, the table reports the impact of Δp_{s-1}^{LND}, the return in London before the departure of a boat, capturing the impact of the arrival of public news. Robust bootstrapped (1,000 replications) standard errors are reported in parentheses. *** and ** denote statistical significance at the 1% and 5% level, respectively.

	Amsterdam Boat Returns ($\Delta p_t^{AMS.boat}$)			Amsterdam No-Boat Returns ($\Delta p_{t+d}^{AMS.no=boat}$)		
	EIC	BoE	3% ann.	EIC	BoE	3% ann.
London postdeparture	0.233	0.138	0.147	0.138	0.168	0.132
returns (Δp_s^{LND})	(0.042)***	(0.050)***	(0.067)**	(0.034)***	(0.043)***	(0.052)***
London predeparture	0.373	0.405	0.503			
returns (Δp_{s-1}^{LND})	(0.042)***	(0.054)***	(0.070)***			
Constant	0.045	0.025	0.033	0.006	0.003	0.006
	(0.028)	(0.017)	(0.022)	(–0.024)	(–0.016)	(0.020)
N	622	640	669	467	465	479

Table XII: Private Information, News and No-News Returns

6.8 Tables XIII and XIV: trading costs and the variance decomposition

Read:

- Table XIII estimates order-handling and inventory-cost components. Trading-cost variance is essentially zero for the EIC but meaningful for SSC, BoE, and the annuities.
- Table XIV is the main decomposition table. For boat returns, public news explains 54% of EIC variance, 56% of BoE variance, and 48% of 3% annuity variance.
- For no-boat returns, private information explains 39% for EIC, 39% for BoE, and 26% for 3% annuities. Trading costs explain 38% of BoE no-boat variance and 54% of 3% annuity no-boat variance.
- Overall, the model explains most return variance for BoE and 3% annuities. The EIC still has a large residual component, especially on no-boat days.

Economic message: The paper's headline result is a decomposition: public news is big, but private information and liquidity/trading frictions together account for a large share of short-run volatility.

Table XIII

Trading Costs

This table presents estimates of trading costs. Panel A refers to order handling costs. The "fees" indicate official brokers' fees. Fees were 0.125% of a security's nominal or par value. To arrive at the fees presented here, the "nominal" fee is multiplied by the average security price. Estimates of Roll's (1984b) "raw" trading cost measure are given by $c = -\sqrt{\text{cov}(\Delta p_t, \Delta p_{t-1})}$. For the EIC, the raw Roll measure cannot be calculated since the first-order autocovariance is positive. The panel therefore also presents estimates using Hasbrouck's (2009) Bayesian estimation. The implied variance is given by $\text{var}(\Delta p_t) = 2c^2$. Panel (B) refers to inventory costs and estimates are based on the simple model presented in the main text: $\Delta p_t = \Delta f_t + \lambda q_t$, where $q_t = \{1, -1\}$ with probability $\{\pi_t, 1 - \pi_t\}$, where $\pi_t = \min\left\{\max[1/2 + \rho(f_{t-1} - p_{t-1}), 1], 0\right\}$. Estimates of λ and ρ are based on minimizing the mean squared error of the empirical and model-implied autocovariances (up to 12 lags; four weeks). Section V.F in the Internet Appendix provides a graphical presentation of the fit of the data and the parameters that minimize the mean squared error. The implied variance is calculated from the simulated return series.

	EIC	SSC	BoE	3% ann.	4% ann.
Panel A: Order Handling Costs					
Fees (%)	0.078	0.142	0.090	0.159	0.142
Roll measure (raw) (%)	–	0.196	0.122	0.225	0.192
Implied variance	0	0.066	0.048	0.100	0.082
Roll measure (Hasbrouck) (%)	0.047	0.101	0.054	0.111	0.099
Implied variance	0.004	0.020	0.006	0.025	0.020
Panel B: Inventory Costs					
λ	0	0.263	0.222	0.323	0.283
ρ	0	3.111	1.899	2.869	1.455
Implied variance	0	0.069	0.049	0.105	0.080

Table XIII: Trading Costs

Table XIV

Summary Results

This table presents results of a simple variance decomposition. The impact of public and private information is identified through the covariances with London pre- and postdeparture returns, Δp_s^{LND} and Δp_{s-1}^{LND}:

$$\text{var}\left(\Delta p_t^{AMS,news}\right) = \text{cov}\left(\Delta p_t^{AMS,news}, \Delta p_{s-1}^{LND}\right) + \text{cov}\left(\Delta p_t^{AMS,news}, \Delta p_s^{LND}\right) + \sigma_{w_t}^2, \text{ and}$$

$$\text{var}\left(\Delta p_{t+d}^{AMS,no=news}\right) = \text{cov}\left(\Delta p_{t+d}^{AMS,no=news}, \Delta p_s^{LND}\right) + \sigma_{w_{t+d}}^2$$

(see Section IV.D for details). The impact of trading costs is identified through simulations of a simple inventory cost model. See Section IV.E and Table XIII for details.

	EIC		BoE		3% ann.	
	$\text{var}(\Delta p_t^{AMS})$	% total	$\text{var}(\Delta p_t^{AMS})$	% total	$\text{var}(\Delta p_t^{AMS})$	% total
Boat returns	0.703		0.229		0.425	
Attributed to:						
Public news	0.380	54.0%	0.129	56.4%	0.202	47.5%
Private information	0.277	39.3%	0.045	19.5%	0.057	13.4%
Trading costs	0	0%	0.049	21.6%	0.105	24.6%
Residual	0.047	6.6%	0.006	2.5%	0.061	14.4%
N	681		681		681	
No-boat returns	0.279		0.131		0.194	
Attributed to:						
Private information	0.110	39.4%	0.051	39.3%	0.051	26.3%
Trading costs	0	0%	0.049	37.7%	0.105	53.9%
Residual	0.169	60.6%	0.030	23.0%	0.038	19.8%
N	481		481		481	
All returns	0.528		0.188		0.330	
Attributed to:						
Public news	0.223	42.2%	0.076	40.2%	0.114	35.9%
Private information	0.208	39.4%	0.047	25.3%	0.057	16.5%
Trading costs	0	0%	0.049	26.1%	0.105	31.8%
Residual	0.097	18.5%	0.016	8.5%	0.047	15.8%

Table XIV: Summary Results

6.9 Table XV: why the EIC is different

Read:

- Table XV studies whether return reversals depend on whether the previous and current periods involved boat arrivals.
- For the EIC, reversals appear mainly for no-boat returns and on subsequent news days.
- This pattern is consistent with liquidity trades clustering in no-boat periods and reversing when the next boat resolves uncertainty.
- The paper also notes an alternative interpretation: sentiment, rumors, or differences of opinion may have moved EIC prices during no-news periods.

Economic message: The EIC's residual no-boat volatility is the paper's main unresolved component. The paper can explain much of volatility generally, but it cannot fully separate adverse-selection-driven liquidity trading from sentiment or disagreement for the EIC.

Table XV

Amsterdam Return Reversals and the Arrival of Boats

This table presents additional details on reversal patterns in security returns. It presents coefficients of the regression $\Delta p_t^{AMS} = \alpha_0 + \alpha_1 \Delta p_{t-1}^{AMS} + u_t$ for different scenarios, indicated by $B_{t-1}, B_t \in \{0, 1\}$, which indicate whether periods t and $t-1$ featured the arrival of a boat. Reported reversal coefficients are based on equations (9) and (10) and sum up the different interaction effects. Differences in coefficients are indicated with Δ. *** and ** indicate whether differences are statistically different at the 1% and 5% level, respectively. The statistical tests are based on robust bootstrapped (1,000 replications) standard errors that come from estimating (9) and (10).

		(1)	(2)	(3)	(4)
		Current return, Δp_t^{AMS}			
		All	$B_t = 1$	$B_t = 0$	Δ
EIC					
Previous return,	$B_{t-1} = 1$	0.058	0.193	−0.023	−0.216**
Δp_{t-1}^{AMS}	$B_{t-1} = 0$	−0.117	−0.160	0.067	0.227**
	Δ	−0.174**	−0.353***	0.090	
BoE					
Previous return,	$B_{t-1} = 1$	−0.074	−0.088	−0.063	0.025
Δp_{t-1}^{AMS}	$B_{t-1} = 0$	−0.051	−0.068	0.070	0.139
	Δ	0.023	0.020	0.133	
3% annuities					
Previous return,	$B_{t-1} = 1$	−0.073	−0.042	−0.095	−0.053
Δp_{t-1}^{AMS}	$B_{t-1} = 0$	−0.277	−0.283	−0.181	0.101
	Δ	−0.204**	−0.241**	−0.087	

Table XV: Amsterdam Return Reversals and the Arrival of Boats

7 Short summary

- The paper studies English securities traded in 18th-century Amsterdam.
- Relevant public information mostly came from London by packet boat.
- Weather delays created observable periods with no new public news from London.
- Amsterdam prices moved strongly when boats arrived.
- But Amsterdam prices also moved substantially when no boats arrived.
- The no-boat/boat variance ratio is roughly 0.40 to 0.58.
- Validation tests show that information mainly flowed from London to Amsterdam during the sample periods.
- Public news explains about 40% of overall volatility.
- Private information explains a large share of both boat and no-boat volatility.
- Local liquidity shocks and trading costs explain another important share, especially for BoE and annuities.
- The EIC has a large unexplained no-boat residual, possibly due to liquidity trades concentrating in no-news periods or to sentiment/disagreement.
- The paper's main contribution is to use a natural interruption of information flows to separate news-driven price changes from trading-driven price changes.

8 Conclusion

8.1 Core conclusion

Koudijs shows that public news is important for short-run price volatility, but it is not the whole story. Amsterdam prices reacted sharply when London news arrived by boat. However, the market also generated substantial volatility during periods with no new public information from London. The evidence suggests that this no-news volatility is largely driven by private information and local liquidity shocks.

8.2 Economic intuition

The paper's economic intuition is simple. If a market remains open while public news is shut down, price changes during the blackout cannot be explained by newly arrived public information. They must come from other forces. In this setting, two forces are central. First, informed traders receive and trade on private signals, gradually revealing information through prices. Second, uninformed liquidity trades move prices temporarily because intermediaries have limited risk-bearing capacity.

8.3 Incremental contribution revisited

The paper advances the volatility literature by replacing noisy modern news measures with directly observed news arrivals. The boats turn information flow into a measurable historical event. This lets the paper compare periods with public news to periods without public news and then use covariance and reversal patterns to identify deeper mechanisms behind price changes.

8.4 Implications for finance

The paper suggests that short-run volatility should not be interpreted mechanically as public-news arrival. Prices may move because private information is incorporated through order flow, because liquidity trades temporarily affect risk premia, or because investors disagree more when objective news is absent. The historical setting therefore speaks to modern debates about price discovery, market microstructure, and excess volatility.

8.5 Limitations

The setting is historical and the securities are concentrated in a small number of English assets traded in Amsterdam. The paper has limited direct trading-volume data. It also cannot fully distinguish between liquidity-driven no-news volatility and sentiment or disagreement for the EIC. Finally, boats carried both public news and private letters, so the decomposition relies on timing and covariance assumptions rather than direct observation of every information signal.

8.6 Takeaway

The enduring takeaway is that news matters, but trading matters too. Even when public information flows stop, markets can continue to generate large price movements. The paper turns a historical communication friction into a clean empirical lens on a central finance question: how much of asset-price volatility is news, and how much is the market process itself?